

Literature Review Plan

Review scope, research questions & source-collection plan

Item	Details
Document	Literature Review Plan
Project phase	Pre-Implementation Phase – Literature Review
Version	v1.0
Date	10/10/2026

1. Purpose

This plan defines what the EgySpy literature review covers, which questions it must answer, and how sources will be found, selected and recorded. Its goal is to keep the three focused reviews aligned with the consolidated proposal (Baseline v1), avoid overlap between review topics, and apply the same method and summary format throughout.

2. Basis: Project Objectives

The review is derived from the objectives in the Consolidated Project Proposal (Baseline v1). They are numbered here only so that research questions and later gap findings can be traced back to them.

ID	Project objective
O1	Track employee movement within the facility in real time.
O2	Record the locations employees visit.
O3	Record the time employees spend at different locations.
O4	Identify movement that may be inconsistent with an employee's assigned work area.
O5	Reduce opportunities for attendance manipulation and fraudulent attendance practices.
O6	Support management monitoring and accountability.

3. Scope

3.1 In scope

- Technologies and systems used to track workforce movement and location inside a defined facility, with **UWB/RTLS as the main focus** and smart attendance systems that use location or movement data.
- Comparison-level coverage of other indoor positioning options (for example BLE, Wi-Fi, RFID) so that the technology choice stays open, as the proposal requires.
- Attacks and fraud methods against attendance and tracking systems (spoofing, relay, device tampering, buddy punching) and the countermeasures documented for them.
- Technologies for transporting and processing location data in real time (latency, scalability, complexity).
- Privacy and employee-monitoring considerations, covered only where they affect system requirements. They are noted within the relevant review topic.

3.2 Out of scope

- Implementation, hardware selection decisions and prototyping. The project is in its planning and design phase only.
- General attendance-management systems (payroll, leave, shift scheduling). Attendance is relevant only in relation to movement after attendance is recorded.
- Evaluating whether an employee is actually performing their assigned work, which the proposal excludes from the system.
- Deep radio-frequency or electronics engineering detail beyond what is needed to compare accuracy and limits.
- Locations outside the defined facility or outside working hours.

3.3 Review principle

Technology and architecture are treated as open. The review informs the later technology and architecture decisions; it does not lock them.

4. Research Questions

ID	Research question	Review topic	Objectives
RQ1	What UWB/RTLS and smart-attendance approaches exist for tracking workforce movement in a facility, and what accuracy, coverage and limitations do they report? How do they compare with other indoor positioning options?	UWB/RTLS and smart attendance	O1, O2, O3
RQ2	How are attendance and tracking systems attacked or manipulated (spoofing, relay, device tampering, buddy punching), and which countermeasures are documented and realistic for a workplace?	Anti-tampering / anti-fraud	O5
RQ3	Which technologies are used to transport and process location data in real time, and how do they compare on latency, scalability and complexity?	Real-time tracking and streaming	O1, O6
RQ4	What needs (including detecting movement inconsistent with an assigned work area) are not fully solved by existing solutions when accurate tracking, fraud resistance and real-time operation are required together?	Comparative synthesis and gap analysis	O4, O5, O6
RQ5	Which findings directly affect requirements and architecture decisions (accuracy targets, security, latency, privacy)?	All review topics	All

5. Source-Collection Plan

5.1 Source types

- Peer-reviewed journal and conference papers, and survey papers.
- Standards and specifications (for example IEEE 802.15.4z for UWB secure ranging).
- Vendor technical documentation and case studies for UWB/RTLS products, used only where they give measurable data.
- Theses and dissertations, as supporting sources.

5.2 Where to search

Google Scholar, IEEE Xplore, ACM Digital Library, ScienceDirect, SpringerLink, MDPI and arXiv. Citation chaining (following the references of key papers and papers that cite them) is used to find further sources.

5.3 Search keywords

Topic	Suggested search strings
UWB/RTLS and attendance	("ultra-wideband" OR UWB OR "real-time location system" OR RTLS) AND ("indoor positioning" OR "employee tracking" OR workforce OR attendance) ("BLE" OR "Bluetooth Low Energy" OR "Wi-Fi" OR RFID) AND "indoor positioning" AND accuracy "smart attendance" AND (IoT OR location OR tracking)
Anti-tampering / anti-fraud	(attendance OR "indoor positioning" OR UWB) AND (spoofing OR "relay attack" OR "buddy punching" OR "proxy attendance" OR tampering) "secure ranging" OR "distance bounding" OR "802.15.4z"
Real-time streaming	("real-time location" OR "location data") AND ("stream processing" OR MQTT OR Kafka OR WebSocket OR "edge computing") AND latency

These strings may be extended during the search. New keywords are added to the shared source log so they can be reused across topics.

5.4 Inclusion and exclusion criteria

Include	Exclude
Credible source with references or verifiable data Published in roughly the last 7 years (older foundational works allowed when no newer equivalent exists) Directly relevant to workforce or facility tracking, or to a clearly applicable threat, countermeasure or technology Reports measurable results (accuracy, latency, detection rate) or clearly states limitations	Articles without references or authorship Marketing material with no technical data Duplicate or near-duplicate sources Sources whose date or origin cannot be verified Sources about outdoor-only or unrelated tracking use cases

5.5 Source summary format

Every source is summarized using the same fields, so the comparative synthesis uses common criteria.

Field	Content
Source ID and citation	Unique ID and full reference in the agreed style
Year and type	Publication year; paper, survey, standard, vendor document, thesis
System / technology / method	What the source studies or proposes
Reported results	Accuracy, latency, detection rate or other measured outcome
Strengths	What it does well
Limitations	Stated limits and limits noticed by the reviewer
Relevance to EgySpy	Link to objectives O1–O6 and any requirement or architecture impact

5.6 Reference management

- One shared reference library (for example a Zotero group) and one shared source log for all review topics.
- One citation style for the whole review. IEEE style is used.
- Findings are written in the team's own words with citations. Copied text is not used.

5.7 Target volume

As a realistic target for the available time, each of the three core reviews aims for about 8–12 relevant sources. The final number depends on the quality and relevance of what is found, and the plan favors fewer strong sources over many weak ones.

6. Review Structure

The review consists of three focused topics followed by one synthesis. Each topic answers one research question and ends with a comparison table and a list of findings that affect requirements or architecture.

Review topic	Focus	Main output
UWB/RTLS workforce tracking and smart attendance systems	Existing systems, technologies, reported accuracy, strengths and limits; comparison with other indoor positioning options	Comparison table of surveyed systems
Anti-tampering / anti-fraud mechanisms	Attack types against attendance and tracking systems and the documented countermeasures	Threat / countermeasure table
Real-time tracking and streaming technologies	Options for transporting and processing location data in real time, compared on latency, scalability and complexity	Option comparison table
Comparative synthesis and gap analysis	The three topics compared on common criteria; unmet needs identified and linked to the project objectives	Comparison table and gap list